

Compound body-state mechanism

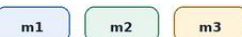
Local primitive plausibility can diverge from aggregate mass properties.

PRIMITIVE COM

Primitive states

each
inert

local terms: m_i, c_i, I_i



locally plausible primitives

PACKAGE COM

Aggregate body

pa
CO

$$M = \sum_i m_i$$



$$c = M^{-1} \sum_i m_i c_i$$

$$I = \sum_i (R I_i R^T + \text{axis term})$$

DIAGNOSTIC GATE

Gate consequence

geometric local plausibility can fail body-state
di

geometry pass → body-state row



ACCEPT

BODY FAIL

What this shows

How COM and inertia aggregation enter the body-level diagnostic path.

What this does not show

It does not claim a general optimal primitive decomposition.